Damien MAÎTRE

MSc Engineering Student — CFD, HPC & Numerical Methods

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Education

Sep 2024 – Jun 2026 **MSc Engineering — Double Degree**

École nationale des ponts et chaussées (ENPC), Paris · Politecnico di Milano, Milan

- Specialisation: computational fluid dynamics, structural mechanics, HPC
- Relevant courses: Turbulence modelling, Numerical methods for PDEs, Finite element methods, High-performance scientific computing
- Graduation research project (*PFE*): [brief topic, e.g. URANS simulation of bluff-body wake at Re $\approx 6 \times 10^8$ using OpenFOAM]

Sep 2021 – Jun 2024 **Bachelor & first-year MSc — *Classes préparatoires* then Engineering School**

[Preparatory school name], [City]

- Intensive programme in mathematics, physics and engineering sciences
- Ranked [X/N] at entrance examination

Research & Engineering Projects

2025 – 2026 **CFD Simulation of a Large-Scale Ellipsoidal Body**

Personal / graduation project — OpenFOAM, gmsh, HPC cluster

- Designed full CFD pipeline: geometry, boundary-layer meshing (gmsh), URANS 3-D simulation at Re $\approx 6 \times 10^8$, GCI mesh-independence study
- Vortex-shedding analysis: Strouhal number extraction, spectral post-processing
- Implemented a custom `BoundaryCorner` field in C++ within a forked gmsh codebase to resolve degenerate mesh elements at axis singularities
- HPC strategy: MPI scaling benchmark, AmgX pressure solver (NVIDIA A100)

2025 – 2026 **Structural FEM & Eurocode Verification — Aluminium Airship Frame**

Personal project — OpenSeesPy, Python

- Eigenvalue buckling analysis (α_{cr}), natural frequency extraction, Rayleigh damping calibration
- Eurocode 9 (EN 1999-1-1) section verification pipeline in Python; aeroelastic assessment (Scruton number, VIV lock-in)
- CFD-to-structural load transfer: conservative force mapping from OpenFOAM pressure fields to FEM nodes

[Year] **[Other academic project title]**

[Course or lab] — [tools used]

Key contribution or result
Method or tool developed

Professional Experience

[Month–Month Year] **[Job Title] — [Company Name]**

Rail engineering / Infrastructure, [City]

come — quantify where possible
nodology, or stakeholder context

[Month–Month Year] **[Job Title] — [Company Name]**

[Sector], [City]

Concrete task and outcome

Technical Skills

CFD & FEM OpenFOAM (RANS/URANS, pimpleFoam, simpleFoam), StarCCM+, Abaqus, OpenSeesPy

Meshing	gmsh (custom C++ fork), Pointwise
HPC & Dev	MPI, OpenMP, AmgX, Linux/UNIX (10+ yrs), Git, Bash, CMake
Programming	Python (NumPy, SciPy, Matplotlib), C++, MATLAB, LaTeX
Post-processing	ParaView, Tecplot, FFT/spectral analysis, VTK pipelines

Languages

French — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B2) · **German** — Working proficiency (B2)

Activities & Community Involvement

[Years] **Rescuer — French Red Cross**
Volunteer first-aid responder

[Years] **Director — Student Theatre / Ensemble**
[Association name], [City]

ers, productions, budget if notable

[Years] **General Secretary — Student Association**
[Association name], [University]

or achievement worth mentioning